

The Figure-Eight Knot

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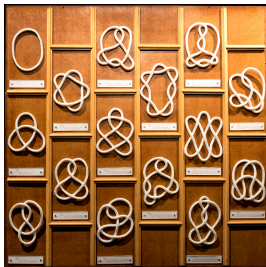
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- #### 4 Decomposition of Figure-Eight Knot

What is a knot?

Think of twisting a string in the real world, and then connecting the endpoints.



Definition

Knot

A **knot** is defined by an injective function from S^1 into $\mathbb{R}^3$.

Knot Projections

A **knot projection** is a particular drawing of the knot (i.e. how we represent it).

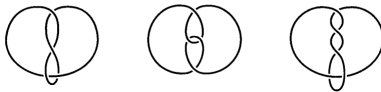


Figure 1.5 Three projections of the figure-eight knot.

Some motivating questions

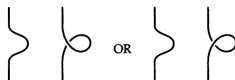
Knot Equivalence

Two projections are equivalent if one can be rearranged into the other without passing through itself (ambient isotopy).

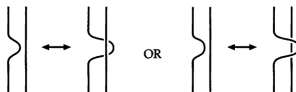


The question becomes, given two knot presentations, how can we determine if they are the same knot?

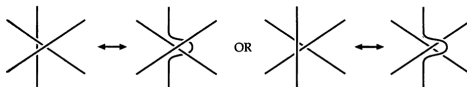
Reidemeister Moves Visualized



(Type I) *Figure 1.22* Type I Reidemeister move.



(Type II) *Figure 1.23* Type II Reidemeister move.



(Type III) *Figure 1.24* Type III Reidemeister move.

These give us a way to manipulate knot projections and should mesh well with our intuitive understanding of unknotting.

Some Invariants

Some invariants give us ways to check if knot projections are not equivalent.

Unknotting Number

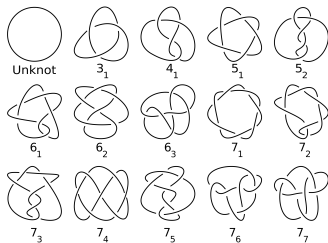
The **unknotting number** of a knot K is the minimal number of changes of the crossings of K to turn it into the unknot over all projections of K .



Some Invariants Pt. 2

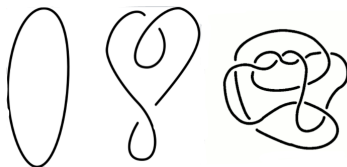
Crossing Number

The **crossing number** of a knot K is the least number of crossings across all projections of K .



However, all of these knots come from a combinatorial approach.
This poses some issues:

- 1 Reidemeister moves are inefficient and can be tedious to deal with
- 2 The invariants we've seen require us to check them over all possible projections of a knot. This is difficult in general.



Classification of Knots

Theorem (Classification of Knots)

Every knot is a torus, satellite, or hyperbolic knot.



Torus knots



Satellite knots

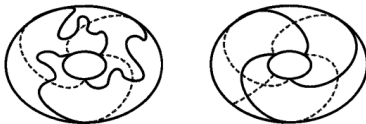


Hyperbolic knots

Torus

Torus

Torus knots are knots which can be drawn to lie on a torus, without any crossings of the knot.

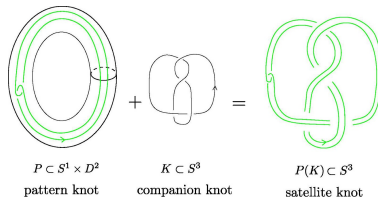


Two pictures of the same torus knot.

Satellite

Satellite

Let K_1 be a non-trivial knot inside a solid torus. We can imagine twisting the torus in some manner defined by another companion knot K_2 . Then, the result K_3 is called a **satellite knot**.

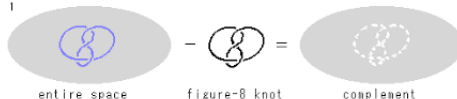


Knot Complements

Idea

Instead of studying the knot K , we can associate it with a **knot complement**, which is $S^3 - K$.

Fig. 1



Uniqueness of Knot Complements

Gordon-Luecke Theorem (1989)

If the complements of two knots are homeomorphic, then the knots are equivalent.

In particular, this means two distinct knots cannot have the same knot complement. This allows us to develop this association between a knot and its complement.

Hyperbolic Space

There are many equivalent definition, but we use the Poincaré half-space model.

Hyperbolic Space

Hyperbolic 3-space H^3 is the upper half of R^3 and a point at infinity equipped with a norm

$$\|p\| = \frac{x^2 + y^2 + z^2}{z^2}$$

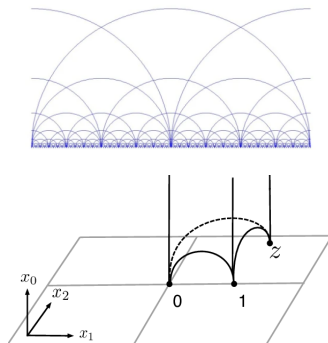
and distance metric induced by the norm

$$d(p, q) = \|p - q\| = \frac{(p_x - q_x)^2 + (p_y - q_y)^2 + (p_z - q_z)^2}{(p_z - q_z)^2}$$

From this, we can define geometric concepts.

Visualizing Hyperbolic Space

- 1 Points higher up "appear" closer.
- 2 Shortest paths are geodesics.
- 3 Tetrahedron and triangles are formed by geodesics.

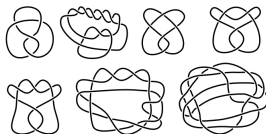


Hyperbolic Knots

Hyperbolic Knots

A knot K is said to be **hyperbolic** if $S^3 - K$ admits a complete metric with curvature -1 .

In particular, this means its knot complement can be equipped with the metric previously mentioned.



Our goal is to decompose this knot complement into smaller pieces to study.

Ideal Shapes

Polyhedron

A **polyhedron** is a closed 3-ball, whose boundary can be labeled with a finite graph, corresponding to edges and vertices, with faces being simply connected.

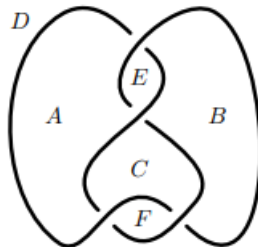
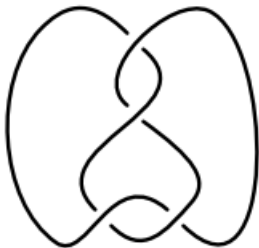
Ideal Polyhedron

An **ideal polyhedron** is a polyhedron without the vertices.

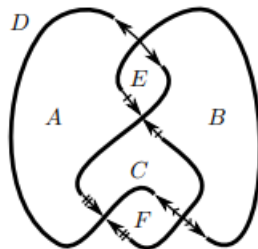
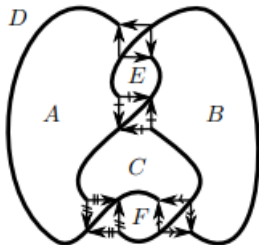
Why ideal polyhedron? When we take the knot complement, K is "one-dimensional" in the sense it is a string of points.

We will demonstrate decomposition using the figure-eight knot as an example.

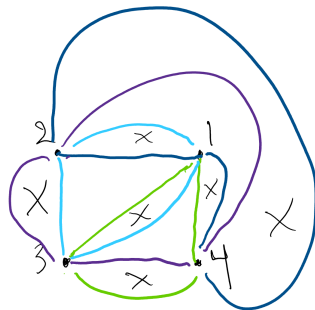
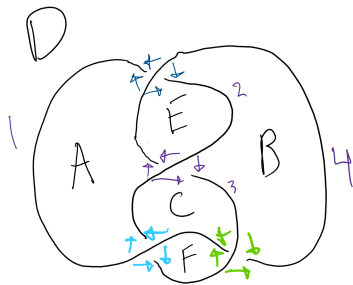
Step 1: Label faces and "arcs"



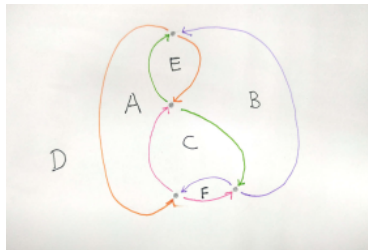
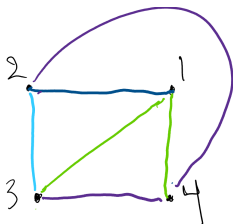
Step 1: Label faces and "arcs"



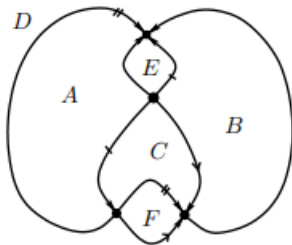
Step 2: Create associated graph



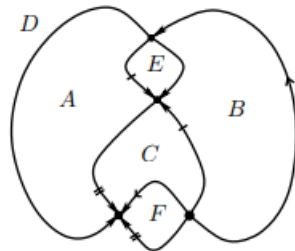
Step 2: Create associated graph



Step 3: Repeat for "back" of knot

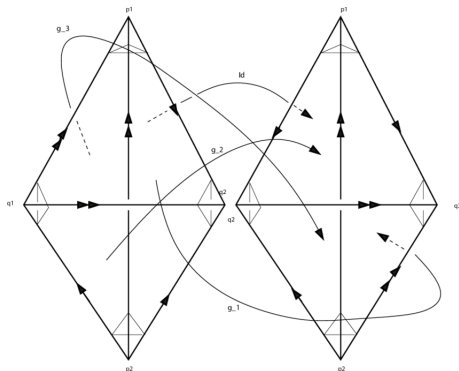


On the left is the top diagram.



On the right is the bottom diagram.

Step 4: Glue them together



Tessellation of Hyperbolic Space

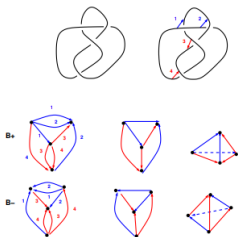
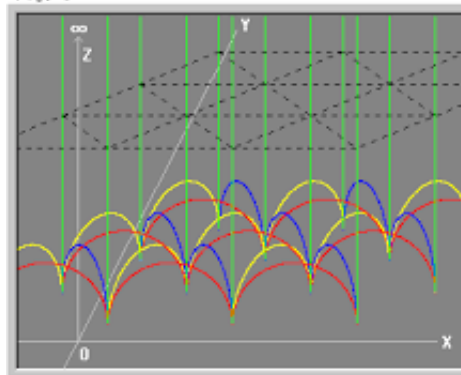


Fig. 9



Conclusion

- ① This generalizes for much, more complicated knots and represents an algorithm that a computer can run to decompose the knot complement into ideal polyhedron. This can help with determining if a knot is hyperbolic.
- ② It turns out that hyperbolic space can help us study knot complements and assign new invariants.

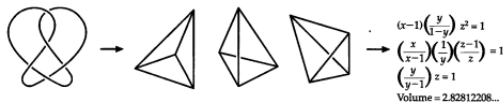


Figure 5.31 Finding the hyperbolic volume of a knot.

Credits

Thanks to the MATH Directed Reading Program for this opportunity, and especially, to my mentor, Yi Wang, for guidance throughout the semester.

References

- ① The Knot Book by Colin Adams
- ② Topology of Surfaces by L. Christine Kinsey
- ③ Hyperbolic Knot Theory by Jessica Purcell
- ④ Algebraic Topology by Allen Hatcher